8 Use your calculator to work out $\sqrt{\dfrac{\sin 25° + \sin 40°}{\cos 25° - \cos 40°}}$

(a) Write down all the figures on your calculator display.

Use degree mode and retain the outer square root.

sin 25° + sin 40° = 1.065405871...

cos 25° − cos 40° = 0.140263344...

Square root of their quotient: 2.756039570...

1 mark
Calculate the numerator and denominator

2.756039570

1 mark
Correct calculator value; keep all displayed figures

(2)

(b) Write your answer to part (a) correct to 2 decimal places.

The third decimal digit is 6, so round 2.75 up.

2.76

1 mark
Correct to 2 decimal places; follow-through allowed

(1)

(Total for Question 8 is 3 marks)